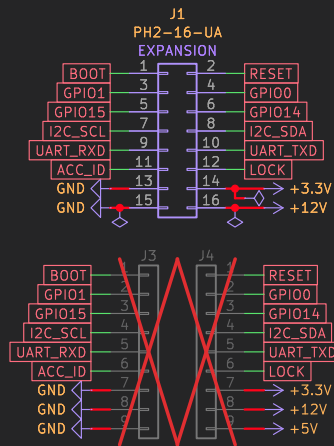


EXPANSION PORT

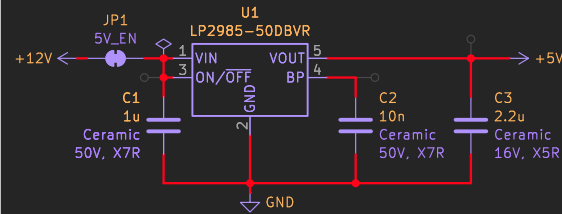
Header for accessories and factory programming

- GPIO0, GPIO1, GPIO14, GPIO15 can be used for any purpose
- UART_RXD and UART_TXD provide the serial port
- I2C_SCL and I2C_SDA provide the I2C bus (QWIIC)
- RESET and BOOT are wired in parallel with their corresponding buttons (active low)
- LOCK engages the safety lock (active low)
- ACC_ID identifies the expansion port accessory
- 12 V supply is unregulated, 1 A current available
- 3.3 V supply is regulated, 600 mA current available



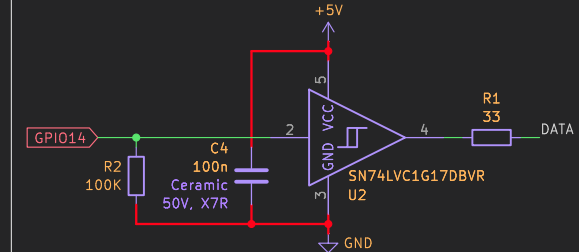
5 V POWER SUPPLY

Supplies 150 mA, ultra low quiescent current (microamps)



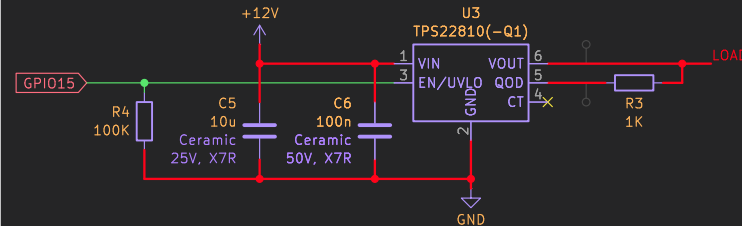
5 V LEVEL SHIFTER FOR ADDRESSABLE LED DATA

The driver can supply up to 32 mA at 5 V
33 ohm resistor reduces signal reflections and protects the output



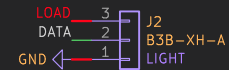
HIGH-SIDE SWITCH FOR 1 A LOAD AT 12 V

Not intended to drive PWM loads



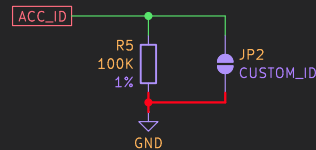
OUTPUT CONNECTOR

Plug the addressable LEDs in here



ACCESSORY IDENTIFICATION

By default, this circuit board identifies itself to Minuet as a "light accessory" when plugged into the expansion port. If you repurpose this circuit board for a different function, close the 'CUSTOM ACC ID' solder jumper to prevent the firmware from automatically configuring the expansion GPIOs.



github.com/brown-studios/minuet

Brown Studios LLC

Sheet: /

File: light.kicad_sch

Title: Minuet Light Accessory

Size: A4 Date: 2026-04

KiCad E.D.A. 9.0.6

Rev: v4.0

Id: 1/1